

Lecture 8

August 12, 2026

W04-L02

k -Nearest Neighbour Estimator

$$\hat{f}_k(x) = \frac{\sum_{i: x_i \in N_k(x)} Y_i}{k}$$

For the large sample consistency of this estimator we need $k \rightarrow \infty$ and $k/n \rightarrow 0$ as $n \rightarrow \infty$. This is a piecewise constant function; not continuous.

Instead of k -NN, one can think about using a weighted k -NN estimator. This is given by

$$\hat{f}_{k\text{-WNN}}(x) = \frac{\sum_{i: x_i \in N_k(x)} \omega(\|x - x_i\|) Y_i}{\sum_{i: x_i \in N_k(x)} \omega(\|x - x_i\|)}$$

where $\omega : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a non-increasing / decreasing function.

Even if $\omega(\cdot)$ is a continuous function, $\hat{f}_{k\text{-WNN}}$ may not be a continuous function unless $k = n$. For $k = n$, we have the following expression for the estimator

$$\hat{f}(x) = \frac{\sum_{i=1}^n \omega(\|x - x_i\|) Y_i}{\sum_{i=1}^n \omega(\|x - x_i\|)}$$

This is continuous if $\omega(\cdot)$ is a continuous function.

Choice of ω : ω should be decreasing / non-increasing function.

One popular choice is the Gaussian weight function

$$\omega(t) = e^{-\frac{1}{2}t^2}$$

For this choice of the weight function, we have

$$\hat{f}(x) = \frac{\sum_{i=1}^n e^{-\frac{1}{2}\|x-x_i\|^2} Y_i}{\sum_{i=1}^n e^{-\frac{1}{2}\|x-x_i\|^2}}$$

In order to make adjustments for the scale, one can think of using an estimator of the form

$$\hat{f}(x) = \frac{\sum_{i=1}^n e^{-\frac{1}{2h^2}\|x-x_i\|^2} Y_i}{\sum_{i=1}^n e^{-\frac{1}{2h^2}\|x-x_i\|^2}} = \frac{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)}$$

where

$$K(t) = e^{-\frac{1}{2}t't} = e^{-\frac{1}{2}\|t\|^2}.$$

Instead of using $K(t) = e^{-\frac{1}{2}t't}$, one can use other kernel functions as well.

This type of estimator is known as the kernel regression estimator or the Nadaraya–Watson estimator.

$K(\cdot)$ is called the kernel function and h is the associated smoothing parameter known as the bandwidth.

Nadaraya–Watson Estimator

NW Est.:

$$\hat{f}_h(x) = \frac{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)}$$

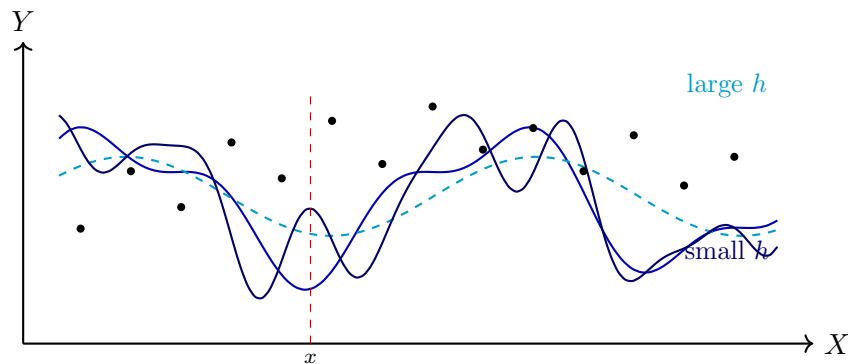


Illustration of the effect of bandwidth h : large h yields a smoother estimate; small h follows the data more closely.

Assume that

$$K(b) = e^{-\frac{1}{2}b^2}.$$

Let $x_1, x_2 \in \mathbb{R}^p$ and $x \in \mathbb{R}^p$.

$$\frac{K\left(\frac{x-x_1}{h}\right)}{K\left(\frac{x-x_2}{h}\right)}.$$

If $\|x - x_1\| < \|x - x_2\|$, then

$$\frac{K\left(\frac{x-x_1}{h}\right)}{K\left(\frac{x-x_2}{h}\right)} = \frac{e^{-\frac{1}{2h^2}\|x-x_1\|^2}}{e^{-\frac{1}{2h^2}\|x-x_2\|^2}} = e^{-\frac{1}{2h^2}\{\|x-x_1\|^2 - \|x-x_2\|^2\}} \uparrow \infty \quad \text{as } h \downarrow 0.$$

Or in other words,

$$\frac{K\left(\frac{x-x_2}{h}\right)}{K\left(\frac{x-x_1}{h}\right)} \downarrow 0 \quad \text{as } h \downarrow 0.$$

Data points: $x_1, \dots, x_n$

Query point: x

Let x_k be the nearest neighbour of x .

$$\lim_{h \downarrow 0} \hat{f}_h(x) = \frac{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)} = Y_k.$$

Cross-validation method can be used to choose the value of h .

Data

$$\{x_1, x_2, \dots, x_n\}, \quad \{Y_1, Y_2, \dots, Y_n\}.$$

$$\min_a \sum (Y_i - a)^2 \longrightarrow a = \bar{Y}.$$

$$\hat{f}(x) = a, \quad \min_a \sum (Y_i - a)^2 K\left(\frac{x-x_i}{h}\right).$$

If we minimise $\sum (Y_i - a)^2$ w.r.t. a , we get $\hat{a} = \bar{Y}$.

If we minimise $\sum (Y_i - a)^2 K\left(\frac{x-x_i}{h}\right)$ w.r.t. a , we get

$$\hat{a} = \frac{\sum K\left(\frac{x-x_i}{h}\right) Y_i}{\sum K\left(\frac{x-x_i}{h}\right)} \longrightarrow \text{Kernel regression estimator (Nadaraya-Watson estimator)}.$$

Locally Constant and Local Linear Estimators

Because of this the Nadaraya-Watson estimator is often called the **locally constant estimator**.

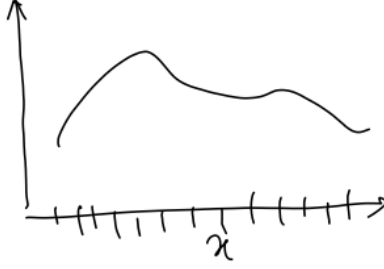
How to get a local linear estimator?

For finding the local linear estimate at x , we shall minimise

$$\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 K\left(\frac{x - X_i}{h}\right) \quad (\text{in the univariate case}).$$

Let $(\hat{\beta}_0, \hat{\beta}_1)$ be the minimiser. Then the local linear estimate of the regression function at x is given by

$$\hat{f}_{LL,h}(x) = \hat{\beta}_0 + \hat{\beta}_1 x.$$



Local linear estimation (alternative method)

We assume that the regression function is linear near x :

$$Y = \beta_0 + \beta_1 X = Y_0 + Y_1(X - x).$$

Minimise

$$\sum_{i=1}^n (Y_i - Y_0 - Y_1(X_i - x))^2 K\left(\frac{x - X_i}{h}\right).$$

Let $(\hat{Y}_0, \hat{Y}_1)$ be the minimiser. Then

$$\hat{Y} \text{ at } x = \hat{Y}_0.$$

Local Polynomial Regression

For finding $\hat{f}(x)$, we minimise

$$\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i - \beta_2 X_i^2 - \cdots - \beta_p X_i^p)^2 K\left(\frac{x - X_i}{h}\right)$$

to get $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$.

The local polynomial estimate of f at x is given by

$$\hat{f}_{LP}(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \cdots + \hat{\beta}_p x^p.$$

Alternative method

Minimise

$$\sum_{i=1}^n \left\{ Y_i - Y_0 - Y_1(X_i - x) - Y_2(X_i - x)^2 - \cdots - Y_p(X_i - x)^p \right\}^2 K\left(\frac{x - X_i}{h}\right).$$

If $(\hat{Y}_0, \hat{Y}_1, \dots, \hat{Y}_p)$ is the minimiser, the local polynomial estimate of f at x is given by

$$\hat{f}_{\text{LP},h}(x) = \hat{Y}_0.$$